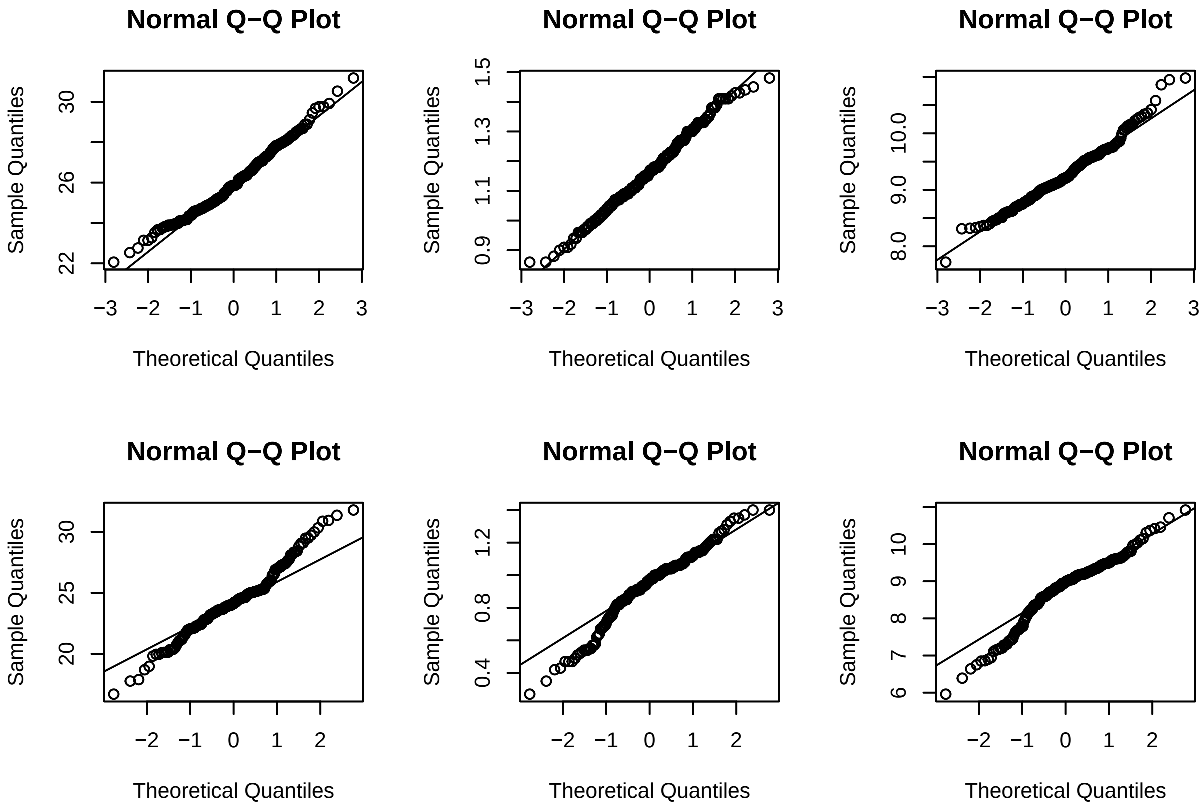
# Untitled

masrir

2026-06-07

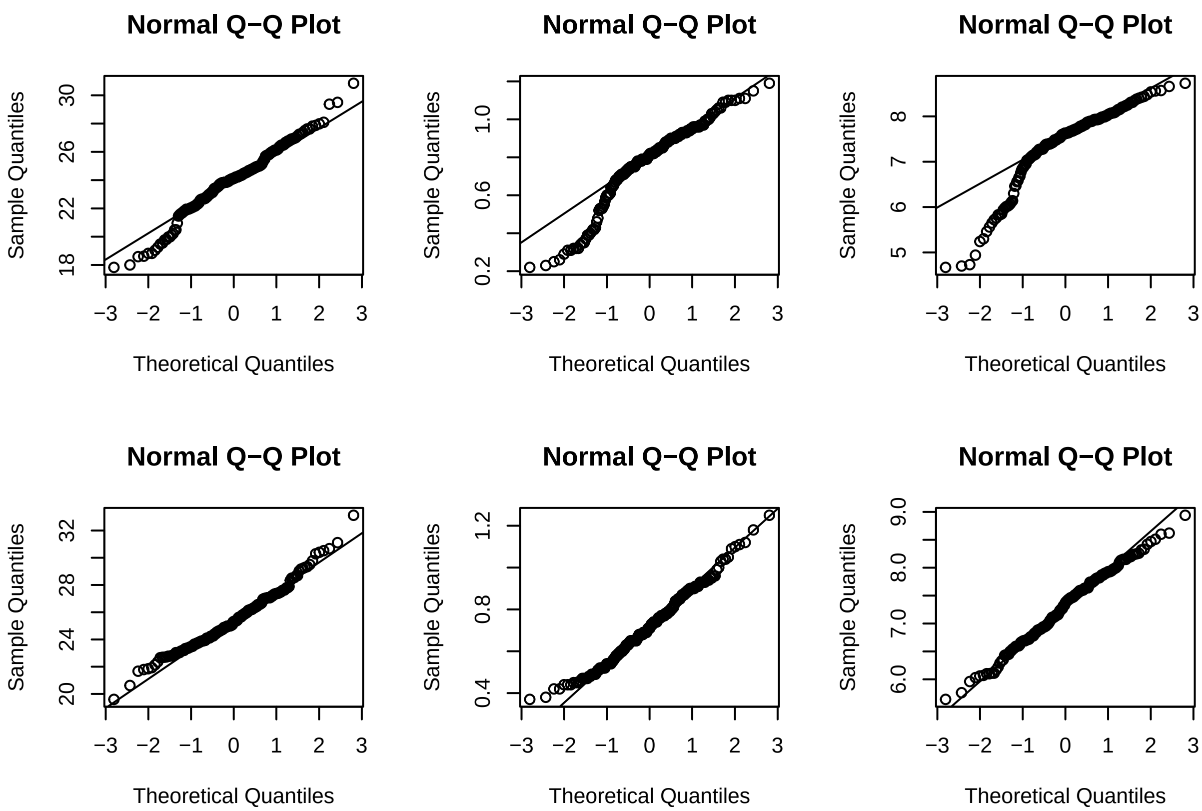
```
#Check for normality (majdool)
df<- read.csv("~/Desktop/Book1.csv")
par (mfrow= c(2,3))
qqnorm(df$length[df$cultivar=="majdool" & df$quality=="high"])
qqline (df$length[df$cultivar=="majdool" & df$quality=="high"])
qqnorm(df$weight[df$cultivar=="majdool" & df$quality=="high"])
qqline (df$weight[df$cultivar=="majdool" & df$quality=="high"])
qqnorm(df$width[df$cultivar=="majdool" & df$quality=="high"])
qqline (df$width[df$cultivar=="majdool" & df$quality=="high"])

qqnorm(df$length[df$cultivar=="majdool" & df$quality=="low"])
qqline (df$length[df$cultivar=="majdool" & df$quality=="low"])
qqnorm(df$weight[df$cultivar=="majdool" & df$quality=="low"])
qqline (df$weight[df$cultivar=="majdool" & df$quality=="low"])
qqnorm(df$width[df$cultivar=="majdool" & df$quality=="low"])
qqline (df$width[df$cultivar=="majdool" & df$quality=="low"])
```



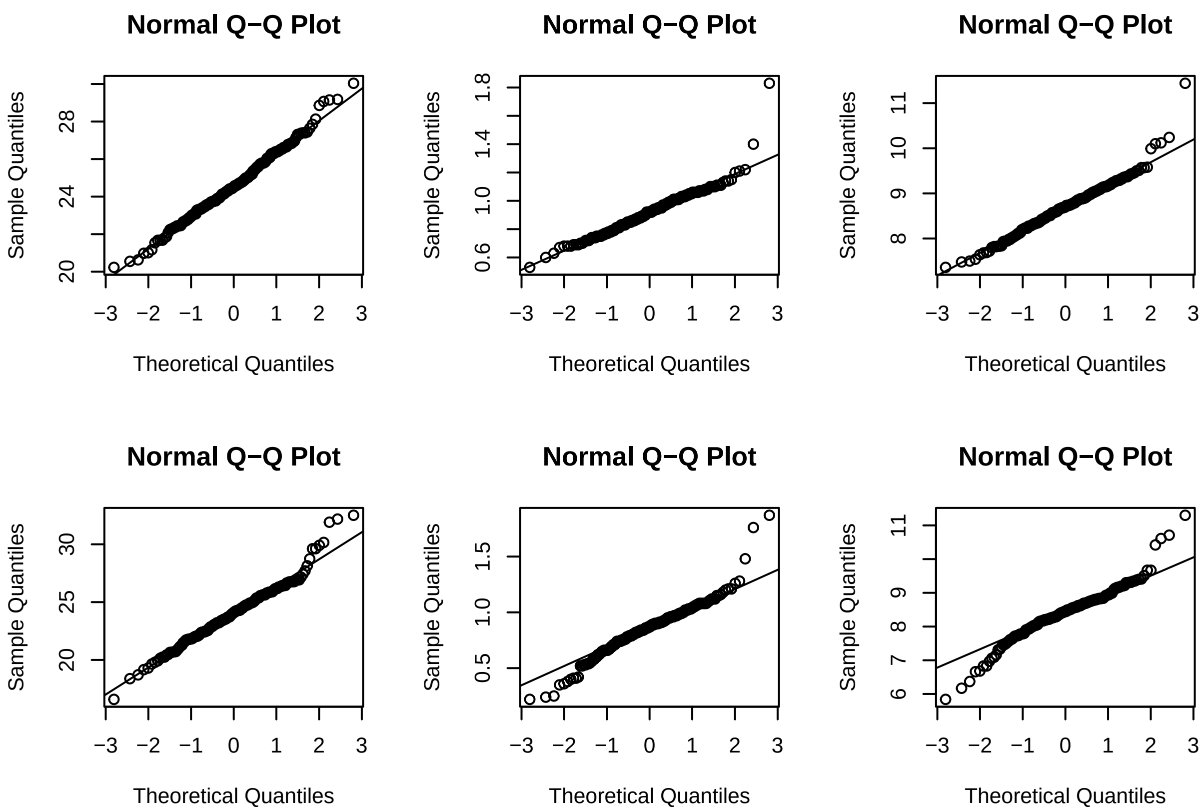

```
#Check for normality (khudri)
par (mfrow= c(2,3))
qqnorm(df$length[df$cultivar=="khudri" & df$quality=="high"])
qqline (df$length[df$cultivar=="khudri" & df$quality=="high"])
qqnorm(df$weight[df$cultivar=="khudri" & df$quality=="high"])
qqline (df$weight[df$cultivar=="khudri" & df$quality=="high"])
qqnorm(df$width[df$cultivar=="khudri" & df$quality=="high"])
qqline (df$width[df$cultivar=="khudri" & df$quality=="high"])

qqnorm(df$length[df$cultivar=="khudri" & df$quality=="low"])
qqline (df$length[df$cultivar=="khudri" & df$quality=="low"])
qqnorm(df$weight[df$cultivar=="khudri" & df$quality=="low"])
qqline (df$weight[df$cultivar=="khudri" & df$quality=="low"])
qqnorm(df$width[df$cultivar=="khudri" & df$quality=="low"])
qqline (df$width[df$cultivar=="khudri" & df$quality=="low"])
```




```
#Check for normality (saqi)
par (mfrow= c(2,3))
qqnorm(df$length[df$cultivar=="saqi" & df$quality=="high"])
qqline (df$length[df$cultivar=="saqi" & df$quality=="high"])
qqnorm(df$weight[df$cultivar=="saqi" & df$quality=="high"])
qqline (df$weight[df$cultivar=="saqi" & df$quality=="high"])
qqnorm(df$width[df$cultivar=="saqi" & df$quality=="high"])
qqline (df$width[df$cultivar=="saqi" & df$quality=="high"])

qqnorm(df$length[df$cultivar=="saqi" & df$quality=="low"])
qqline (df$length[df$cultivar=="saqi" & df$quality=="low"])
qqnorm(df$weight[df$cultivar=="saqi" & df$quality=="low"])
qqline (df$weight[df$cultivar=="saqi" & df$quality=="low"])
qqnorm(df$width[df$cultivar=="saqi" & df$quality=="low"])
qqline (df$width[df$cultivar=="saqi" & df$quality=="low"])
```




Assumption of normality is approved

```r
library(car)
```

```
## Loading required package: carData
```

```r
#Khudri
khudri <- subset(df, cultivar == "khudri")

leveneTest(width ~ quality, data = khudri)
```

```
## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.2275 0.6337
##       397
```

```r
leveneTest(weight ~ quality, data=khudri)
```

```
## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.8437 0.3589
##       397
```

```r
leveneTest(length ~ quality, data= khudri)
```

```
## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.0464 0.8295
##       397
```

p-values » alpha, which means that H0 of equal variances of Khudri is not violated.

```r
#Majdool
majdool <- subset(df, cultivar == "majdool")

leveneTest(width ~ quality, data = majdool)
```

```
## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value    Pr(>F)
## group   1  22.571 2.897e-06 ***
##       372
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
leveneTest(weight ~ quality, data=majdool)
```

```
## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value    Pr(>F)
## group   1  26.456 4.369e-07 ***
##       372
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
leveneTest(length ~ quality, data=majdool)
```

```
## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value    Pr(>F)
## group   1  19.787 1.145e-05 ***
##       372
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

For Majdool, Levene's test indicated unequal variances for length, width, and weight (all p < 0.001); therefore, Welch's t-tests were used for these comparisons in t-tests

```r
#Saqi
saqi<- subset(df, cultivar == "saqi")

leveneTest(width ~ quality, data = saqi)
```

```
## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value  Pr(>F)
## group   1  4.9218 0.02708 *
##       398
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
leveneTest(weight ~ quality, data=saqi)
```

```
## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value    Pr(>F)
## group   1  13.987 0.0002111 ***
##       398
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
leveneTest(length ~ quality, data= saqi)
```

```
## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value    Pr(>F)
## group   1  13.042 0.0003435 ***
##       398
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

For Saqi, Levene's test indicated unequal variances for length, width, and weight (all p < 0.001); therefore, Welch's t-tests were used for these comparisons in t-tests

```r
df <- read.csv("~/Desktop/Book1.csv")

base_colors <- c(
  "khudri"  = "#009E73",
  "majdool" = "#E69F00",
  "saqi"    = "#56B4E9"
)

colors <- c(
  adjustcolor(base_colors["khudri"],  alpha.f = 0.7),
  adjustcolor(base_colors["khudri"],  alpha.f = 1.0),
  adjustcolor(base_colors["majdool"], alpha.f = 0.7),
  adjustcolor(base_colors["majdool"], alpha.f = 1.0),
  adjustcolor(base_colors["saqi"],    alpha.f = 0.7),
  adjustcolor(base_colors["saqi"],    alpha.f = 1.0)
)

df$cultivar <- tolower(trimws(df$cultivar))
df$quality  <- tolower(trimws(df$quality))

df$group <- paste(df$cultivar, df$quality, sep = " ")
df$group <- factor(df$group, levels = c(
  "khudri low",
  "khudri high",
  "majdool low",
  "majdool high",
  "saqi low",
  "saqi high"
))

positions <- c(1, 2, 4, 5, 7, 8)

format_p_label <- function(p) {
  if (is.character(p)) return(p)
  if (p < 0.001) {
    format(p, scientific = TRUE, digits = 3)
  } else {
    format(round(p, 3), nsmall = 3)
  }
}

add_sig <- function(x1, x2, y, h, p, text_offset, cex = 0.68, lwd = 1) {
  segments(x1, y,     x1, y + h, lwd = lwd, xpd = FALSE)
  segments(x1, y + h, x2, y + h, lwd = lwd, xpd = FALSE)
  segments(x2, y + h, x2, y,     lwd = lwd, xpd = FALSE)

  label <- if (is.character(p) && grepl("^<", p)) {
    paste0("p ", p)
  } else {
    paste0("p = ", format_p_label(p))
  }

  text(
```

```
    x = mean(c(x1, x2)),
    y = y + h + text_offset,
    labels = label,
    cex = cex,
    xpd = FALSE
  )
}

plot_one <- function(y, ylab, main_title, pvals, y_data_top, y_ticks) {
  y_max <- max(y, na.rm = TRUE)
  y_min <- min(y, na.rm = TRUE)
  y_range <- y_max - y_min

  if (y_range == 0) {
    y_range <- max(abs(y_max), 1) * 0.2
  }

  base_y      <- y_max + y_range * 0.015
  step_y      <- y_range * 0.08
  bracket_h   <- y_range * 0.02
  text_offset <- y_range * 0.04

  y_annot_top <- base_y + (length(pvals) - 1) * step_y + bracket_h + text_offset + y_range * 0.03
  ylim_top <- max(y_data_top, y_annot_top)

  boxplot(
    y ~ df$group,
    at = positions,
    col = colors,
    outline = FALSE,
    las = 2,
    main = main_title,
    xlab = "",
    ylab = ylab,
    ylim = c(0, ylim_top),
    yaxt = "n",
    names = c("Low", "High", "Low", "High", "Low", "High"),
    cex.axis = 0.9
  )

  axis(2, at = y_ticks, labels = y_ticks, las = 1)

  x1 <- c(1, 4, 7)
  x2 <- c(2, 5, 8)

  for (i in seq_along(pvals)) {
    add_sig(
      x1 = x1[i],
      x2 = x2[i],
      y = base_y + (i - 1) * step_y,
      h = bracket_h,
      p = pvals[i],
      text_offset = text_offset
```

```r
    )
  }

  axis(
    1,
    at = c(1.5, 4.5, 7.5),
    labels = c("Khudri", "Majdool", "Saqi"),
    tick = FALSE,
    line = 4
  )
}

p_length <- c(4.2e-11, 1.004e-12, 0.009)
p_weight <- c(0.002, "< 2.2e-16", 0.001)
p_width  <- c(0.08, 8.12e-10, 8.53e-06)

png("/Users/Masrir/Figure4_1.png", width = 18, height = 11, units = "cm", res = 300)

par(
  mfrow = c(1, 3),
  mar = c(10, 5, 4.5, 2) + 0.1
)

plot_one(
  df$length,
  "Seed Length (mm)",
  "Seed Length",
  p_length,
  y_data_top = 32,
  y_ticks = seq(0, 30, by = 5)
)

plot_one(
  df$weight,
  "Seed Weight (g)",
  "Seed Weight",
  p_weight,
  y_data_top = 2.0,
  y_ticks = seq(0, 2.0, by = 0.5)
)

plot_one(
  df$width,
  "Seed Width (mm)",
  "Seed Width",
  p_width,
  y_data_top = 10,
  y_ticks = seq(0, 10, by = 2)
)

dev.off()
```

```
## pdf
```

```
##   2
```

```r
#T-test for Khudri(length)
##H0: Low and High quality Majdool seeds'length have equal means
##H1: Low and High quality Majdool seeds' length have different means
t.test(length~quality, data= subset(df, cultivar=="khudri"))
```

```
##
##  Welch Two Sample t-test
##
## data:  length by quality
## t = -6.7867, df = 394.55, p-value = 4.222e-11
## alternative hypothesis: true difference in means between group high and group low is not equal to 0
## 95 percent confidence interval:
##  -1.885272 -1.038343
## sample estimates:
## mean in group high  mean in group low
##           24.00849           25.47030
```

For 98% confidence interval, p-value< 0.02 which means that H0 is rejected. Low quality Khudri seeds are longer

```r
#T-test for Majdool (length)
##H0: Low and High quality Majdool seeds'length have equal means
##H1: Low and High quality Majdool seeds' length have different means
t.test(length~quality, data= subset(df, cultivar=="majdool"), var.equal= FALSE)
```

```
##
##  Welch Two Sample t-test
##
## data:  length by quality
## t = 7.4712, df = 281.65, p-value = 1.004e-12
## alternative hypothesis: true difference in means between group high and group low is not equal to 0
## 95 percent confidence interval:
##  1.280337 2.196319
## sample estimates:
## mean in group high  mean in group low
##           26.01844           24.28011
```

For 98% confidence interval, p-value< 0.02 which means that H0 is rejected. High quality Majdool seeds are longer

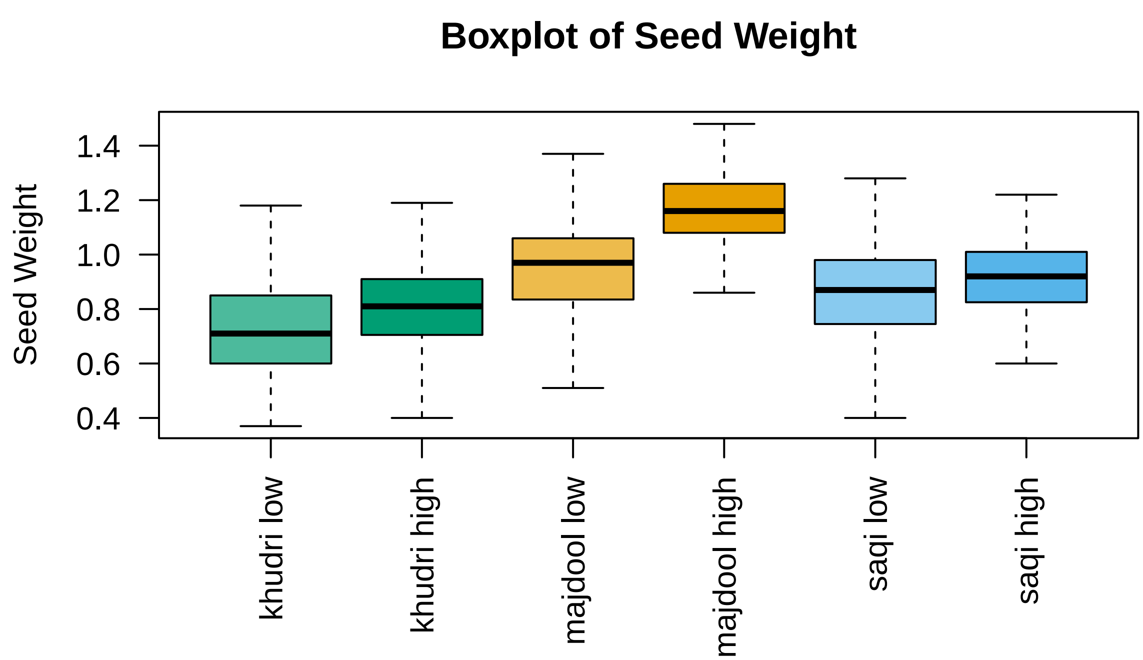
```r
#T-test for Saqi (length)
##H0: Low and High quality Saqi seeds'length have equal means
##H1: Low and High quality Saqi seeds' length have different means
t.test(length~quality, data= subset(df, cultivar=="saqi"), var.equal=FALSE)
```

```
##
##  Welch Two Sample t-test
##
## data:  length by quality
```

```
## t = 2.6152, df = 361.4, p-value = 0.00929
## alternative hypothesis: true difference in means between group high and group low is not equal to 0
## 95 percent confidence interval:
##  0.1369687 0.9674313
## sample estimates:
## mean in group high  mean in group low
##            24.6126            24.0604
```

For 98% confidence interval, p-value< 0.02 which means that H0 is rejected. High quality Saqi seeds are longer

```r
par(mar=c(10, 5, 4, 2))
boxplot(df$weight ~ df$group,
        main="Boxplot of Seed Weight",
     xlab="",
        ylab="Seed Weight",
        las=2, col= colors, outline = FALSE)
```




```r
#T-test for Khudri (weight)
##H0: Low and High quality Khudri seeds'weight have equal means
##H1: Low and High quality Khudri seeds' weight have different means
t.test(weight~quality, data= subset(df, cultivar=="khudri"))
```

```
##
##  Welch Two Sample t-test
```

```
##
## data:  weight by quality
## t = 3.013, df = 385.65, p-value = 0.002758
## alternative hypothesis: true difference in means between group high and group low is not equal to 0
## 95 percent confidence interval:
##  0.01960988 0.09327052
## sample estimates:
## mean in group high  mean in group low
##          0.7780402          0.7216000
```

For 98% confidence interval, p-value$<$ 0.02 which means that H0 is rejected. High quality Khudri seeds are heavier

```r
#T-test for Majdool (weight)
##H0: Low and High quality Majdool seeds'weight have equal means
##H1: Low and High quality Majdool seeds'weight have different means
t.test(weight~quality, data= subset(df, cultivar=="majdool"), var.equal=FALSE)
```

```
##
##  Welch Two Sample t-test
##
## data:  weight by quality
## t = 12.303, df = 276.26, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group high and group low is not equal to 0
## 95 percent confidence interval:
##  0.1960327 0.2707154
## sample estimates:
## mean in group high  mean in group low
##          1.1662312          0.9328571
```

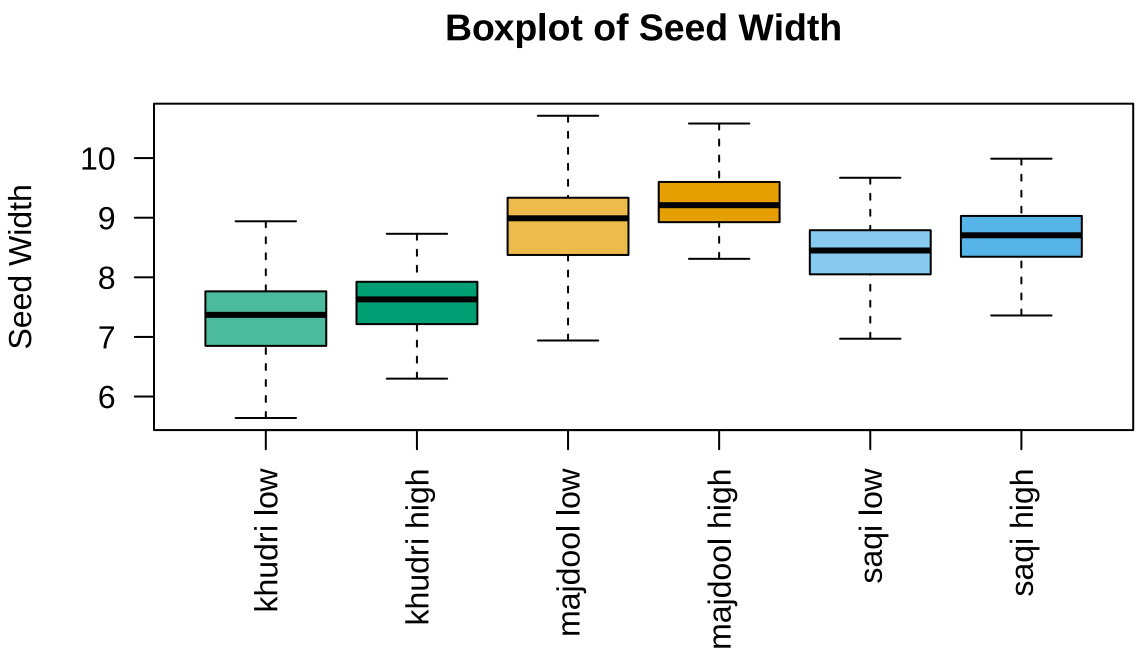
For 98% confidence interval, p-value$<$ 0.02 which means that H0 is rejected. High quality Majdool seeds are heavier

```r
#T-test for Saqi (weight)
##H0: Low and High quality Saqi seeds' weight have equal means
##H1: Low and High quality Saqi seeds' weight have different means
t.test(weight~quality, data= subset(df, cultivar=="saqi"), var.equal=FALSE)
```

```
##
##  Welch Two Sample t-test
##
## data:  weight by quality
## t = 3.1584, df = 346.53, p-value = 0.001726
## alternative hypothesis: true difference in means between group high and group low is not equal to 0
## 95 percent confidence interval:
##  0.02252347 0.09687653
## sample estimates:
## mean in group high  mean in group low
##             0.9195             0.8598
```

For 98% confidence interval, p-value$<$ 0.02 which means that H0 is rejected. High quality Saqi seeds are heavier

```r
par(mar=c(10, 5, 4, 2))
boxplot(df$width ~ df$group,
        main="Boxplot of Seed Width",
     xlab="",
        ylab="Seed Width",
        las=2, col= colors, outline = FALSE)
```




```r
#T-test for Khudri (width)
##H0: Low and High quality Khudri seeds'width have equal means
##H1: Low and High quality Khudri seeds' width have different means
t.test(width~quality, data= subset(df, cultivar=="khudri"))
```

```
##
##  Welch Two Sample t-test
##
## data:  width by quality
## t = 1.7118, df = 377.22, p-value = 0.08776
## alternative hypothesis: true difference in means between group high and group low is not equal to 0
## 95 percent confidence interval:
##  -0.01810828  0.26172939
## sample estimates:
## mean in group high  mean in group low
##           7.422111           7.300300
```

For 98% confidence interval, p-value> 0.02 which means that H0 is not rejected and that there is no statistical significance between low and high quality seeds' widths

```r
#T-test for Majdool (width)
##H0: Low and High quality Majdool seeds'width have equal means
##H1: Low and High quality Majdool seeds' width have different means
t.test(width~quality, data= subset(df, cultivar=="majdool"), var.equal=FALSE)
```

```
##
##  Welch Two Sample t-test
##
## data:  width by quality
## t = 6.3629, df = 277.97, p-value = 8.128e-10
## alternative hypothesis: true difference in means between group high and group low is not equal to 0
## 95 percent confidence interval:
##  0.3347750 0.6347119
## sample estimates:
## mean in group high  mean in group low
##           9.275829           8.791086
```

For 98% confidence interval, p-value< 0.02 which means that H0 is rejected. High quality Majdool seeds are wider

```r
#T-test for Saqi (width)
##H0: Low and High quality Saqi seeds'width have equal means
##H1: Low and High quality Saqi seeds' width have different means
t.test(width~quality, data= subset(df, cultivar=="saqi"), var.equal=FALSE)
```

```
##
##  Welch Two Sample t-test
##
## data:  width by quality
## t = 4.5144, df = 372.34, p-value = 8.532e-06
## alternative hypothesis: true difference in means between group high and group low is not equal to 0
## 95 percent confidence interval:
##  0.1622994 0.4128006
## sample estimates:
## mean in group high  mean in group low
##            8.69630            8.40875
```

For 98% confidence interval, p-value< 0.02 which means that H0 is rejected. High quality Saqi seeds are wider

```r
getwd()
```

```
## [1] "/Users/Masrir"
```